

NCERT CLASS 8

60 Questions Per Topic

Based on the refined 2026–27 curriculum map created for this project.

Each curriculum topic below contains exactly 60 questions: understanding, examples/classification, comparison, application, reasoning/error analysis, and higher-order activity.

These are question-bank items derived from the topic descriptions in the supplied curriculum map. Validate against the exact school/textbook edition before high-stakes use.

MATHEMATICS — 1. A Square and A Cube

Coverage: ■ Perfect squares/cubes; patterns; square roots; cube roots; factorisation-based methods.

Understanding

1. What is ■ Perfect squares/cubes, and what are its main features? Use a simple example.
2. What is patterns, and what are its main features? Use a school-life example.
3. What is square roots, and what are its main features? Use a real-world example.
4. What is cube roots, and what are its main features? Show the idea with a diagram where appropriate.
5. What is factorisation-based methods., and what are its main features? Explain in your own words.
6. What is ■ Perfect squares/cubes, and what are its main features? Give a step-by-step response.
7. What is patterns, and what are its main features? Mention one common misconception.
8. What is square roots, and what are its main features? State one practical application.
9. What is cube roots, and what are its main features? Justify your response.
10. What is factorisation-based methods., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Perfect squares/cubes and explain each. Use a simple example.
12. Give two examples related to patterns and explain each. Use a school-life example.
13. Give two examples related to square roots and explain each. Use a real-world example.
14. Give two examples related to cube roots and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to factorisation-based methods. and explain each. Explain in your own words.
16. Give two examples related to ■ Perfect squares/cubes and explain each. Give a step-by-step response.
17. Give two examples related to patterns and explain each. Mention one common misconception.
18. Give two examples related to square roots and explain each. State one practical application.
19. Give two examples related to cube roots and explain each. Justify your response.
20. Give two examples related to factorisation-based methods. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Perfect squares/cubes related to patterns? Compare the two ideas. Use a simple example.
22. How is patterns related to square roots? Compare the two ideas. Use a school-life example.
23. How is square roots related to cube roots? Compare the two ideas. Use a real-world example.
24. How is cube roots related to factorisation-based methods.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is factorisation-based methods. related to ■ Perfect squares/cubes? Compare the two ideas. Explain in your own words.
26. How is ■ Perfect squares/cubes related to patterns? Compare the two ideas. Give a step-by-step response.
27. How is patterns related to square roots? Compare the two ideas. Mention one common misconception.
28. How is square roots related to cube roots? Compare the two ideas. State one practical application.
29. How is cube roots related to factorisation-based methods.? Compare the two ideas. Justify your response.
30. How is factorisation-based methods. related to ■ Perfect squares/cubes? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Perfect squares/cubes to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply patterns to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply square roots to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply cube roots to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply factorisation-based methods. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Perfect squares/cubes to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply patterns to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply square roots to a realistic situation. What would you do or calculate? State one practical application.
39. Apply cube roots to a realistic situation. What would you do or calculate? Justify your response.
40. Apply factorisation-based methods. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Perfect squares/cubes. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of patterns. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of square roots. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of cube roots. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of factorisation-based methods.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Perfect squares/cubes. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of patterns. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of square roots. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of cube roots. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of factorisation-based methods.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Perfect squares/cubes. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show patterns. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show square roots. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show cube roots. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show factorisation-based methods.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Perfect squares/cubes. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show patterns. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show square roots. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show cube roots. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show factorisation-based methods.. Include the relevant rule or principle.

MATHEMATICS — 2. Power Play

Coverage: ■ Exponents; laws of exponents; negative exponents; standard form/scientific notation.

Understanding

1. What is ■ Exponents, and what are its main features? Use a simple example.
2. What is laws of exponents, and what are its main features? Use a school-life example.
3. What is negative exponents, and what are its main features? Use a real-world example.
4. What is standard form/scientific notation., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Exponents, and what are its main features? Explain in your own words.
6. What is laws of exponents, and what are its main features? Give a step-by-step response.
7. What is negative exponents, and what are its main features? Mention one common misconception.
8. What is standard form/scientific notation., and what are its main features? State one practical application.
9. What is ■ Exponents, and what are its main features? Justify your response.
10. What is laws of exponents, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to negative exponents and explain each. Use a simple example.
12. Give two examples related to standard form/scientific notation. and explain each. Use a school-life example.
13. Give two examples related to ■ Exponents and explain each. Use a real-world example.
14. Give two examples related to laws of exponents and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to negative exponents and explain each. Explain in your own words.
16. Give two examples related to standard form/scientific notation. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Exponents and explain each. Mention one common misconception.
18. Give two examples related to laws of exponents and explain each. State one practical application.
19. Give two examples related to negative exponents and explain each. Justify your response.
20. Give two examples related to standard form/scientific notation. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Exponents related to laws of exponents? Compare the two ideas. Use a simple example.
22. How is laws of exponents related to negative exponents? Compare the two ideas. Use a school-life example.
23. How is negative exponents related to standard form/scientific notation.? Compare the two ideas. Use a real-world example.
24. How is standard form/scientific notation. related to ■ Exponents? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Exponents related to laws of exponents? Compare the two ideas. Explain in your own words.
26. How is laws of exponents related to negative exponents? Compare the two ideas. Give a step-by-step response.
27. How is negative exponents related to standard form/scientific notation.? Compare the two ideas. Mention one common misconception.
28. How is standard form/scientific notation. related to ■ Exponents? Compare the two ideas. State one practical application.
29. How is ■ Exponents related to laws of exponents? Compare the two ideas. Justify your response.
30. How is laws of exponents related to negative exponents? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply negative exponents to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply standard form/scientific notation. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Exponents to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply laws of exponents to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply negative exponents to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply standard form/scientific notation. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Exponents to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply laws of exponents to a realistic situation. What would you do or calculate? State one practical application.
39. Apply negative exponents to a realistic situation. What would you do or calculate? Justify your response.
40. Apply standard form/scientific notation. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Exponents. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of laws of exponents. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of negative exponents. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of standard form/scientific notation.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Exponents. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of laws of exponents. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of negative exponents. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of standard form/scientific notation.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Exponents. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of laws of exponents. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show negative exponents. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show standard form/scientific notation.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Exponents. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show laws of exponents. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show negative exponents. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show standard form/scientific notation.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Exponents. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show laws of exponents. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show negative exponents. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show standard form/scientific notation.. Include the relevant rule or principle.

MATHEMATICS — 3. A Story of Numbers

Coverage: ■ Evolution of numerals and number systems; place-value ideas; mathematical history.

Understanding

1. What is ■ Evolution of numerals and number systems, and what are its main features? Use a simple example.
2. What is place-value ideas, and what are its main features? Use a school-life example.
3. What is mathematical history., and what are its main features? Use a real-world example.
4. What is ■ Evolution of numerals and number systems, and what are its main features? Show the idea with a diagram where appropriate.
5. What is place-value ideas, and what are its main features? Explain in your own words.
6. What is mathematical history., and what are its main features? Give a step-by-step response.
7. What is ■ Evolution of numerals and number systems, and what are its main features? Mention one common misconception.
8. What is place-value ideas, and what are its main features? State one practical application.
9. What is mathematical history., and what are its main features? Justify your response.
10. What is ■ Evolution of numerals and number systems, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to place-value ideas and explain each. Use a simple example.
12. Give two examples related to mathematical history. and explain each. Use a school-life example.
13. Give two examples related to ■ Evolution of numerals and number systems and explain each. Use a real-world example.
14. Give two examples related to place-value ideas and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to mathematical history. and explain each. Explain in your own words.
16. Give two examples related to ■ Evolution of numerals and number systems and explain each. Give a step-by-step response.
17. Give two examples related to place-value ideas and explain each. Mention one common misconception.
18. Give two examples related to mathematical history. and explain each. State one practical application.
19. Give two examples related to ■ Evolution of numerals and number systems and explain each. Justify your response.
20. Give two examples related to place-value ideas and explain each. Include the relevant rule or principle.

Comparison

21. How is mathematical history. related to ■ Evolution of numerals and number systems? Compare the two ideas. Use a simple example.
22. How is ■ Evolution of numerals and number systems related to place-value ideas? Compare the two ideas. Use a school-life example.
23. How is place-value ideas related to mathematical history.? Compare the two ideas. Use a real-world example.
24. How is mathematical history. related to ■ Evolution of numerals and number systems? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Evolution of numerals and number systems related to place-value ideas? Compare the two ideas. Explain in your own words.
26. How is place-value ideas related to mathematical history.? Compare the two ideas. Give a step-by-step response.
27. How is mathematical history. related to ■ Evolution of numerals and number systems? Compare the two ideas. Mention one common misconception.
28. How is ■ Evolution of numerals and number systems related to place-value ideas? Compare the two ideas. State one practical application.
29. How is place-value ideas related to mathematical history.? Compare the two ideas. Justify your response.
30. How is mathematical history. related to ■ Evolution of numerals and number systems? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Evolution of numerals and number systems to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply place-value ideas to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply mathematical history. to a realistic situation. What would you do or calculate? Use a real-world example.

34. Apply ■ Evolution of numerals and number systems to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply place-value ideas to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply mathematical history. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Evolution of numerals and number systems to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply place-value ideas to a realistic situation. What would you do or calculate? State one practical application.
39. Apply mathematical history. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Evolution of numerals and number systems to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of place-value ideas. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of mathematical history.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Evolution of numerals and number systems. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of place-value ideas. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of mathematical history.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Evolution of numerals and number systems. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of place-value ideas. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of mathematical history.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Evolution of numerals and number systems. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of place-value ideas. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show mathematical history.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Evolution of numerals and number systems. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show place-value ideas. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show mathematical history.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Evolution of numerals and number systems. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show place-value ideas. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show mathematical history.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Evolution of numerals and number systems. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show place-value ideas. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show mathematical history.. Include the relevant rule or principle.

MATHEMATICS — 4. Quadrilaterals

Coverage: ■ Types; sides/angles; properties; diagonals; geometric reasoning.

Understanding

1. What is ■ Types, and what are its main features? Use a simple example.
2. What is sides/angles, and what are its main features? Use a school-life example.
3. What is properties, and what are its main features? Use a real-world example.
4. What is diagonals, and what are its main features? Show the idea with a diagram where appropriate.
5. What is geometric reasoning., and what are its main features? Explain in your own words.
6. What is ■ Types, and what are its main features? Give a step-by-step response.
7. What is sides/angles, and what are its main features? Mention one common misconception.
8. What is properties, and what are its main features? State one practical application.
9. What is diagonals, and what are its main features? Justify your response.
10. What is geometric reasoning., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Types and explain each. Use a simple example.
12. Give two examples related to sides/angles and explain each. Use a school-life example.
13. Give two examples related to properties and explain each. Use a real-world example.
14. Give two examples related to diagonals and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to geometric reasoning. and explain each. Explain in your own words.
16. Give two examples related to ■ Types and explain each. Give a step-by-step response.
17. Give two examples related to sides/angles and explain each. Mention one common misconception.
18. Give two examples related to properties and explain each. State one practical application.
19. Give two examples related to diagonals and explain each. Justify your response.
20. Give two examples related to geometric reasoning. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Types related to sides/angles? Compare the two ideas. Use a simple example.
22. How is sides/angles related to properties? Compare the two ideas. Use a school-life example.
23. How is properties related to diagonals? Compare the two ideas. Use a real-world example.
24. How is diagonals related to geometric reasoning.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is geometric reasoning. related to ■ Types? Compare the two ideas. Explain in your own words.
26. How is ■ Types related to sides/angles? Compare the two ideas. Give a step-by-step response.
27. How is sides/angles related to properties? Compare the two ideas. Mention one common misconception.
28. How is properties related to diagonals? Compare the two ideas. State one practical application.
29. How is diagonals related to geometric reasoning.? Compare the two ideas. Justify your response.
30. How is geometric reasoning. related to ■ Types? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Types to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply sides/angles to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply properties to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply diagonals to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply geometric reasoning. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Types to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply sides/angles to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply properties to a realistic situation. What would you do or calculate? State one practical application.
39. Apply diagonals to a realistic situation. What would you do or calculate? Justify your response.
40. Apply geometric reasoning to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Types. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of sides/angles. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of properties. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of diagonals. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of geometric reasoning.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Types. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of sides/angles. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of properties. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of diagonals. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of geometric reasoning.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Types. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show sides/angles. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show properties. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show diagonals. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show geometric reasoning.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Types. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show sides/angles. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show properties. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show diagonals. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show geometric reasoning.. Include the relevant rule or principle.

MATHEMATICS — 5. Number Play

Coverage: ■ Divisibility; number patterns; digit puzzles; logical properties.

Understanding

1. What is ■ Divisibility, and what are its main features? Use a simple example.
2. What is number patterns, and what are its main features? Use a school-life example.
3. What is digit puzzles, and what are its main features? Use a real-world example.
4. What is logical properties., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Divisibility, and what are its main features? Explain in your own words.
6. What is number patterns, and what are its main features? Give a step-by-step response.
7. What is digit puzzles, and what are its main features? Mention one common misconception.
8. What is logical properties., and what are its main features? State one practical application.
9. What is ■ Divisibility, and what are its main features? Justify your response.
10. What is number patterns, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to digit puzzles and explain each. Use a simple example.
12. Give two examples related to logical properties. and explain each. Use a school-life example.
13. Give two examples related to ■ Divisibility and explain each. Use a real-world example.
14. Give two examples related to number patterns and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to digit puzzles and explain each. Explain in your own words.
16. Give two examples related to logical properties. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Divisibility and explain each. Mention one common misconception.
18. Give two examples related to number patterns and explain each. State one practical application.
19. Give two examples related to digit puzzles and explain each. Justify your response.
20. Give two examples related to logical properties. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Divisibility related to number patterns? Compare the two ideas. Use a simple example.
22. How is number patterns related to digit puzzles? Compare the two ideas. Use a school-life example.
23. How is digit puzzles related to logical properties.? Compare the two ideas. Use a real-world example.
24. How is logical properties. related to ■ Divisibility? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Divisibility related to number patterns? Compare the two ideas. Explain in your own words.
26. How is number patterns related to digit puzzles? Compare the two ideas. Give a step-by-step response.
27. How is digit puzzles related to logical properties.? Compare the two ideas. Mention one common misconception.
28. How is logical properties. related to ■ Divisibility? Compare the two ideas. State one practical application.
29. How is ■ Divisibility related to number patterns? Compare the two ideas. Justify your response.
30. How is number patterns related to digit puzzles? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply digit puzzles to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply logical properties. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Divisibility to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply number patterns to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply digit puzzles to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply logical properties. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Divisibility to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply number patterns to a realistic situation. What would you do or calculate? State one practical application.
39. Apply digit puzzles to a realistic situation. What would you do or calculate? Justify your response.
40. Apply logical properties. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Divisibility. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of number patterns. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of digit puzzles. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of logical properties.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Divisibility. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of number patterns. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of digit puzzles. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of logical properties.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Divisibility. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of number patterns. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show digit puzzles. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show logical properties.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Divisibility. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show number patterns. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show digit puzzles. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show logical properties.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Divisibility. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show number patterns. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show digit puzzles. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show logical properties.. Include the relevant rule or principle.

MATHEMATICS — 6. We Distribute, Yet Things Multiply

Coverage: ■ Distributive property; algebraic expressions; expansion; modelling.

Understanding

1. What is ■ Distributive property, and what are its main features? Use a simple example.
2. What is algebraic expressions, and what are its main features? Use a school-life example.
3. What is expansion, and what are its main features? Use a real-world example.
4. What is modelling., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Distributive property, and what are its main features? Explain in your own words.
6. What is algebraic expressions, and what are its main features? Give a step-by-step response.
7. What is expansion, and what are its main features? Mention one common misconception.
8. What is modelling., and what are its main features? State one practical application.
9. What is ■ Distributive property, and what are its main features? Justify your response.
10. What is algebraic expressions, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to expansion and explain each. Use a simple example.
12. Give two examples related to modelling. and explain each. Use a school-life example.
13. Give two examples related to ■ Distributive property and explain each. Use a real-world example.
14. Give two examples related to algebraic expressions and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to expansion and explain each. Explain in your own words.
16. Give two examples related to modelling. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Distributive property and explain each. Mention one common misconception.
18. Give two examples related to algebraic expressions and explain each. State one practical application.
19. Give two examples related to expansion and explain each. Justify your response.
20. Give two examples related to modelling. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Distributive property related to algebraic expressions? Compare the two ideas. Use a simple example.
22. How is algebraic expressions related to expansion? Compare the two ideas. Use a school-life example.
23. How is expansion related to modelling.? Compare the two ideas. Use a real-world example.
24. How is modelling. related to ■ Distributive property? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Distributive property related to algebraic expressions? Compare the two ideas. Explain in your own words.
26. How is algebraic expressions related to expansion? Compare the two ideas. Give a step-by-step response.
27. How is expansion related to modelling.? Compare the two ideas. Mention one common misconception.
28. How is modelling. related to ■ Distributive property? Compare the two ideas. State one practical application.
29. How is ■ Distributive property related to algebraic expressions? Compare the two ideas. Justify your response.
30. How is algebraic expressions related to expansion? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply expansion to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply modelling. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Distributive property to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply algebraic expressions to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply expansion to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply modelling. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Distributive property to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply algebraic expressions to a realistic situation. What would you do or calculate? State one practical application.
39. Apply expansion to a realistic situation. What would you do or calculate? Justify your response.
40. Apply modelling. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Distributive property. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of algebraic expressions. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of expansion. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of modelling.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Distributive property. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of algebraic expressions. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of expansion. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of modelling.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Distributive property. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of algebraic expressions. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show expansion. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show modelling.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Distributive property. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show algebraic expressions. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show expansion. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show modelling.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Distributive property. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show algebraic expressions. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show expansion. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show modelling.. Include the relevant rule or principle.

MATHEMATICS — 7. Proportional Reasoning-1

Coverage: ■ Ratios; rates; proportional relationships; unitary reasoning.

Understanding

1. What is ■ Ratios, and what are its main features? Use a simple example.
2. What is rates, and what are its main features? Use a school-life example.
3. What is proportional relationships, and what are its main features? Use a real-world example.
4. What is unitary reasoning., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Ratios, and what are its main features? Explain in your own words.
6. What is rates, and what are its main features? Give a step-by-step response.
7. What is proportional relationships, and what are its main features? Mention one common misconception.
8. What is unitary reasoning., and what are its main features? State one practical application.
9. What is ■ Ratios, and what are its main features? Justify your response.
10. What is rates, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to proportional relationships and explain each. Use a simple example.
12. Give two examples related to unitary reasoning. and explain each. Use a school-life example.
13. Give two examples related to ■ Ratios and explain each. Use a real-world example.
14. Give two examples related to rates and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to proportional relationships and explain each. Explain in your own words.
16. Give two examples related to unitary reasoning. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Ratios and explain each. Mention one common misconception.
18. Give two examples related to rates and explain each. State one practical application.
19. Give two examples related to proportional relationships and explain each. Justify your response.
20. Give two examples related to unitary reasoning. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Ratios related to rates? Compare the two ideas. Use a simple example.
22. How is rates related to proportional relationships? Compare the two ideas. Use a school-life example.
23. How is proportional relationships related to unitary reasoning.? Compare the two ideas. Use a real-world example.
24. How is unitary reasoning. related to ■ Ratios? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Ratios related to rates? Compare the two ideas. Explain in your own words.
26. How is rates related to proportional relationships? Compare the two ideas. Give a step-by-step response.
27. How is proportional relationships related to unitary reasoning.? Compare the two ideas. Mention one common misconception.
28. How is unitary reasoning. related to ■ Ratios? Compare the two ideas. State one practical application.
29. How is ■ Ratios related to rates? Compare the two ideas. Justify your response.
30. How is rates related to proportional relationships? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply proportional relationships to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply unitary reasoning. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Ratios to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply rates to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply proportional relationships to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply unitary reasoning. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Ratios to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply rates to a realistic situation. What would you do or calculate? State one practical application.
39. Apply proportional relationships to a realistic situation. What would you do or calculate? Justify your response.
40. Apply unitary reasoning. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Ratios. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of rates. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of proportional relationships. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of unitary reasoning.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Ratios. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of rates. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of proportional relationships. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of unitary reasoning.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Ratios. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of rates. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show proportional relationships. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show unitary reasoning.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Ratios. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show rates. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show proportional relationships. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show unitary reasoning.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Ratios. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show rates. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show proportional relationships. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show unitary reasoning.. Include the relevant rule or principle.

MATHEMATICS — 8. Fractions in Disguise

Coverage: ■ Fractions/decimals/percentages; equivalence; applications.

Understanding

1. What is ■ Fractions/decimals/percentages, and what are its main features? Use a simple example.
2. What is equivalence, and what are its main features? Use a school-life example.
3. What is applications., and what are its main features? Use a real-world example.
4. What is ■ Fractions/decimals/percentages, and what are its main features? Show the idea with a diagram where appropriate.
5. What is equivalence, and what are its main features? Explain in your own words.
6. What is applications., and what are its main features? Give a step-by-step response.
7. What is ■ Fractions/decimals/percentages, and what are its main features? Mention one common misconception.
8. What is equivalence, and what are its main features? State one practical application.
9. What is applications., and what are its main features? Justify your response.
10. What is ■ Fractions/decimals/percentages, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to equivalence and explain each. Use a simple example.
12. Give two examples related to applications. and explain each. Use a school-life example.
13. Give two examples related to ■ Fractions/decimals/percentages and explain each. Use a real-world example.
14. Give two examples related to equivalence and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to applications. and explain each. Explain in your own words.
16. Give two examples related to ■ Fractions/decimals/percentages and explain each. Give a step-by-step response.
17. Give two examples related to equivalence and explain each. Mention one common misconception.
18. Give two examples related to applications. and explain each. State one practical application.
19. Give two examples related to ■ Fractions/decimals/percentages and explain each. Justify your response.
20. Give two examples related to equivalence and explain each. Include the relevant rule or principle.

Comparison

21. How is applications. related to ■ Fractions/decimals/percentages? Compare the two ideas. Use a simple example.
22. How is ■ Fractions/decimals/percentages related to equivalence? Compare the two ideas. Use a school-life example.
23. How is equivalence related to applications.? Compare the two ideas. Use a real-world example.
24. How is applications. related to ■ Fractions/decimals/percentages? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Fractions/decimals/percentages related to equivalence? Compare the two ideas. Explain in your own words.
26. How is equivalence related to applications.? Compare the two ideas. Give a step-by-step response.
27. How is applications. related to ■ Fractions/decimals/percentages? Compare the two ideas. Mention one common misconception.
28. How is ■ Fractions/decimals/percentages related to equivalence? Compare the two ideas. State one practical application.
29. How is equivalence related to applications.? Compare the two ideas. Justify your response.
30. How is applications. related to ■ Fractions/decimals/percentages? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Fractions/decimals/percentages to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply equivalence to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply applications. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Fractions/decimals/percentages to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply equivalence to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply applications. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Fractions/decimals/percentages to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply equivalence to a realistic situation. What would you do or calculate? State one practical application.
39. Apply applications. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Fractions/decimals/percentages to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of equivalence. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Fractions/decimals/percentages. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of equivalence. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Fractions/decimals/percentages. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of equivalence. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of applications.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Fractions/decimals/percentages. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of equivalence. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Fractions/decimals/percentages. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show equivalence. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Fractions/decimals/percentages. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show equivalence. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Fractions/decimals/percentages. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show equivalence. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show applications.. Include the relevant rule or principle.

MATHEMATICS — 9. The Baudhayana-Pythagoras Theorem

Coverage: ■ Right triangles; theorem; proof/intuitive verification; applications; Indian mathematical history.

Understanding

1. What is ■ Right triangles, and what are its main features? Use a simple example.
2. What is theorem, and what are its main features? Use a school-life example.
3. What is proof/intuitive verification, and what are its main features? Use a real-world example.
4. What is applications, and what are its main features? Show the idea with a diagram where appropriate.
5. What is Indian mathematical history., and what are its main features? Explain in your own words.
6. What is ■ Right triangles, and what are its main features? Give a step-by-step response.
7. What is theorem, and what are its main features? Mention one common misconception.
8. What is proof/intuitive verification, and what are its main features? State one practical application.
9. What is applications, and what are its main features? Justify your response.
10. What is Indian mathematical history., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Right triangles and explain each. Use a simple example.
12. Give two examples related to theorem and explain each. Use a school-life example.
13. Give two examples related to proof/intuitive verification and explain each. Use a real-world example.
14. Give two examples related to applications and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to Indian mathematical history. and explain each. Explain in your own words.
16. Give two examples related to ■ Right triangles and explain each. Give a step-by-step response.
17. Give two examples related to theorem and explain each. Mention one common misconception.
18. Give two examples related to proof/intuitive verification and explain each. State one practical application.
19. Give two examples related to applications and explain each. Justify your response.
20. Give two examples related to Indian mathematical history. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Right triangles related to theorem? Compare the two ideas. Use a simple example.
22. How is theorem related to proof/intuitive verification? Compare the two ideas. Use a school-life example.
23. How is proof/intuitive verification related to applications? Compare the two ideas. Use a real-world example.
24. How is applications related to Indian mathematical history.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is Indian mathematical history. related to ■ Right triangles? Compare the two ideas. Explain in your own words.
26. How is ■ Right triangles related to theorem? Compare the two ideas. Give a step-by-step response.
27. How is theorem related to proof/intuitive verification? Compare the two ideas. Mention one common misconception.
28. How is proof/intuitive verification related to applications? Compare the two ideas. State one practical application.
29. How is applications related to Indian mathematical history.? Compare the two ideas. Justify your response.
30. How is Indian mathematical history. related to ■ Right triangles? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Right triangles to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply theorem to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply proof/intuitive verification to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply applications to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply Indian mathematical history. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Right triangles to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply theorem to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply proof/intuitive verification to a realistic situation. What would you do or calculate? State one practical application.
39. Apply applications to a realistic situation. What would you do or calculate? Justify your response.
40. Apply Indian mathematical history. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Right triangles. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of theorem. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of proof/intuitive verification. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of applications. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of Indian mathematical history.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Right triangles. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of theorem. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of proof/intuitive verification. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of applications. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of Indian mathematical history.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Right triangles. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show theorem. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show proof/intuitive verification. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show applications. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show Indian mathematical history.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Right triangles. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show theorem. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show proof/intuitive verification. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show applications. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show Indian mathematical history.. Include the relevant rule or principle.

MATHEMATICS — 10. Proportional Reasoning-2

Coverage: ■ Percentages; profit/loss; simple/compound interest; inverse relationships and applications.

Understanding

1. What is ■ Percentages, and what are its main features? Use a simple example.
2. What is profit/loss, and what are its main features? Use a school-life example.
3. What is simple/compound interest, and what are its main features? Use a real-world example.
4. What is inverse relationships and applications., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Percentages, and what are its main features? Explain in your own words.
6. What is profit/loss, and what are its main features? Give a step-by-step response.
7. What is simple/compound interest, and what are its main features? Mention one common misconception.
8. What is inverse relationships and applications., and what are its main features? State one practical application.
9. What is ■ Percentages, and what are its main features? Justify your response.
10. What is profit/loss, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to simple/compound interest and explain each. Use a simple example.
12. Give two examples related to inverse relationships and applications. and explain each. Use a school-life example.
13. Give two examples related to ■ Percentages and explain each. Use a real-world example.
14. Give two examples related to profit/loss and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to simple/compound interest and explain each. Explain in your own words.
16. Give two examples related to inverse relationships and applications. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Percentages and explain each. Mention one common misconception.
18. Give two examples related to profit/loss and explain each. State one practical application.
19. Give two examples related to simple/compound interest and explain each. Justify your response.
20. Give two examples related to inverse relationships and applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Percentages related to profit/loss? Compare the two ideas. Use a simple example.
22. How is profit/loss related to simple/compound interest? Compare the two ideas. Use a school-life example.
23. How is simple/compound interest related to inverse relationships and applications.? Compare the two ideas. Use a real-world example.
24. How is inverse relationships and applications. related to ■ Percentages? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Percentages related to profit/loss? Compare the two ideas. Explain in your own words.
26. How is profit/loss related to simple/compound interest? Compare the two ideas. Give a step-by-step response.
27. How is simple/compound interest related to inverse relationships and applications.? Compare the two ideas. Mention one common misconception.
28. How is inverse relationships and applications. related to ■ Percentages? Compare the two ideas. State one practical application.
29. How is ■ Percentages related to profit/loss? Compare the two ideas. Justify your response.
30. How is profit/loss related to simple/compound interest? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply simple/compound interest to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply inverse relationships and applications. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Percentages to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply profit/loss to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply simple/compound interest to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply inverse relationships and applications. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Percentages to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply profit/loss to a realistic situation. What would you do or calculate? State one practical application.
39. Apply simple/compound interest to a realistic situation. What would you do or calculate? Justify your response.
40. Apply inverse relationships and applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Percentages. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of profit/loss. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of simple/compound interest. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of inverse relationships and applications.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Percentages. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of profit/loss. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of simple/compound interest. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of inverse relationships and applications.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Percentages. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of profit/loss. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show simple/compound interest. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show inverse relationships and applications.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Percentages. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show profit/loss. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show simple/compound interest. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show inverse relationships and applications.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Percentages. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show profit/loss. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show simple/compound interest. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show inverse relationships and applications.. Include the relevant rule or principle.

MATHEMATICS — 11. Exploring Some Geometric Themes

Coverage: ■ Geometric constructions; symmetry; patterns; 3D ideas; nets and visualisation.

Understanding

1. What is ■ Geometric constructions, and what are its main features? Use a simple example.
2. What is symmetry, and what are its main features? Use a school-life example.
3. What is patterns, and what are its main features? Use a real-world example.
4. What is 3D ideas, and what are its main features? Show the idea with a diagram where appropriate.
5. What is nets and visualisation., and what are its main features? Explain in your own words.
6. What is ■ Geometric constructions, and what are its main features? Give a step-by-step response.
7. What is symmetry, and what are its main features? Mention one common misconception.
8. What is patterns, and what are its main features? State one practical application.
9. What is 3D ideas, and what are its main features? Justify your response.
10. What is nets and visualisation., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Geometric constructions and explain each. Use a simple example.
12. Give two examples related to symmetry and explain each. Use a school-life example.
13. Give two examples related to patterns and explain each. Use a real-world example.
14. Give two examples related to 3D ideas and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to nets and visualisation. and explain each. Explain in your own words.
16. Give two examples related to ■ Geometric constructions and explain each. Give a step-by-step response.
17. Give two examples related to symmetry and explain each. Mention one common misconception.
18. Give two examples related to patterns and explain each. State one practical application.
19. Give two examples related to 3D ideas and explain each. Justify your response.
20. Give two examples related to nets and visualisation. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Geometric constructions related to symmetry? Compare the two ideas. Use a simple example.
22. How is symmetry related to patterns? Compare the two ideas. Use a school-life example.
23. How is patterns related to 3D ideas? Compare the two ideas. Use a real-world example.
24. How is 3D ideas related to nets and visualisation.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is nets and visualisation. related to ■ Geometric constructions? Compare the two ideas. Explain in your own words.
26. How is ■ Geometric constructions related to symmetry? Compare the two ideas. Give a step-by-step response.
27. How is symmetry related to patterns? Compare the two ideas. Mention one common misconception.
28. How is patterns related to 3D ideas? Compare the two ideas. State one practical application.
29. How is 3D ideas related to nets and visualisation.? Compare the two ideas. Justify your response.
30. How is nets and visualisation. related to ■ Geometric constructions? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Geometric constructions to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply symmetry to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply patterns to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply 3D ideas to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply nets and visualisation. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Geometric constructions to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply symmetry to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply patterns to a realistic situation. What would you do or calculate? State one practical application.
39. Apply 3D ideas to a realistic situation. What would you do or calculate? Justify your response.
40. Apply nets and visualisation. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Geometric constructions. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of symmetry. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of patterns. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of 3D ideas. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of nets and visualisation.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Geometric constructions. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of symmetry. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of patterns. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of 3D ideas. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of nets and visualisation.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Geometric constructions. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show symmetry. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show patterns. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show 3D ideas. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show nets and visualisation.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Geometric constructions. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show symmetry. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show patterns. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show 3D ideas. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show nets and visualisation.. Include the relevant rule or principle.

MATHEMATICS — 12. Tales by Dots and Lines

Coverage: ■ Coordinate-style representation; mean; line graphs; interpreting trends.

Understanding

1. What is ■ Coordinate-style representation, and what are its main features? Use a simple example.
2. What is mean, and what are its main features? Use a school-life example.
3. What is line graphs, and what are its main features? Use a real-world example.
4. What is interpreting trends., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Coordinate-style representation, and what are its main features? Explain in your own words.
6. What is mean, and what are its main features? Give a step-by-step response.
7. What is line graphs, and what are its main features? Mention one common misconception.
8. What is interpreting trends., and what are its main features? State one practical application.
9. What is ■ Coordinate-style representation, and what are its main features? Justify your response.
10. What is mean, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to line graphs and explain each. Use a simple example.
12. Give two examples related to interpreting trends. and explain each. Use a school-life example.
13. Give two examples related to ■ Coordinate-style representation and explain each. Use a real-world example.
14. Give two examples related to mean and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to line graphs and explain each. Explain in your own words.
16. Give two examples related to interpreting trends. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Coordinate-style representation and explain each. Mention one common misconception.
18. Give two examples related to mean and explain each. State one practical application.
19. Give two examples related to line graphs and explain each. Justify your response.
20. Give two examples related to interpreting trends. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Coordinate-style representation related to mean? Compare the two ideas. Use a simple example.
22. How is mean related to line graphs? Compare the two ideas. Use a school-life example.
23. How is line graphs related to interpreting trends.? Compare the two ideas. Use a real-world example.
24. How is interpreting trends. related to ■ Coordinate-style representation? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Coordinate-style representation related to mean? Compare the two ideas. Explain in your own words.
26. How is mean related to line graphs? Compare the two ideas. Give a step-by-step response.
27. How is line graphs related to interpreting trends.? Compare the two ideas. Mention one common misconception.
28. How is interpreting trends. related to ■ Coordinate-style representation? Compare the two ideas. State one practical application.
29. How is ■ Coordinate-style representation related to mean? Compare the two ideas. Justify your response.
30. How is mean related to line graphs? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply line graphs to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply interpreting trends. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Coordinate-style representation to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply mean to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply line graphs to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply interpreting trends. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Coordinate-style representation to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply mean to a realistic situation. What would you do or calculate? State one practical application.
39. Apply line graphs to a realistic situation. What would you do or calculate? Justify your response.
40. Apply interpreting trends. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Coordinate-style representation. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of mean. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of line graphs. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of interpreting trends.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Coordinate-style representation. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of mean. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of line graphs. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of interpreting trends.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Coordinate-style representation. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of mean. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show line graphs. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show interpreting trends.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Coordinate-style representation. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show mean. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show line graphs. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show interpreting trends.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Coordinate-style representation. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show mean. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show line graphs. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show interpreting trends.. Include the relevant rule or principle.

MATHEMATICS — 13. Algebra Play

Coverage: ■ Algebraic expressions; identities; factorisation; equations and real-life modelling.

Understanding

1. What is ■ Algebraic expressions, and what are its main features? Use a simple example.
2. What is identities, and what are its main features? Use a school-life example.
3. What is factorisation, and what are its main features? Use a real-world example.
4. What is equations and real-life modelling., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Algebraic expressions, and what are its main features? Explain in your own words.
6. What is identities, and what are its main features? Give a step-by-step response.
7. What is factorisation, and what are its main features? Mention one common misconception.
8. What is equations and real-life modelling., and what are its main features? State one practical application.
9. What is ■ Algebraic expressions, and what are its main features? Justify your response.
10. What is identities, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to factorisation and explain each. Use a simple example.
12. Give two examples related to equations and real-life modelling. and explain each. Use a school-life example.
13. Give two examples related to ■ Algebraic expressions and explain each. Use a real-world example.
14. Give two examples related to identities and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to factorisation and explain each. Explain in your own words.
16. Give two examples related to equations and real-life modelling. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Algebraic expressions and explain each. Mention one common misconception.
18. Give two examples related to identities and explain each. State one practical application.
19. Give two examples related to factorisation and explain each. Justify your response.
20. Give two examples related to equations and real-life modelling. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Algebraic expressions related to identities? Compare the two ideas. Use a simple example.
22. How is identities related to factorisation? Compare the two ideas. Use a school-life example.
23. How is factorisation related to equations and real-life modelling.? Compare the two ideas. Use a real-world example.
24. How is equations and real-life modelling. related to ■ Algebraic expressions? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Algebraic expressions related to identities? Compare the two ideas. Explain in your own words.
26. How is identities related to factorisation? Compare the two ideas. Give a step-by-step response.
27. How is factorisation related to equations and real-life modelling.? Compare the two ideas. Mention one common misconception.
28. How is equations and real-life modelling. related to ■ Algebraic expressions? Compare the two ideas. State one practical application.
29. How is ■ Algebraic expressions related to identities? Compare the two ideas. Justify your response.
30. How is identities related to factorisation? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply factorisation to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply equations and real-life modelling. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Algebraic expressions to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply identities to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply factorisation to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply equations and real-life modelling. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Algebraic expressions to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply identities to a realistic situation. What would you do or calculate? State one practical application.
39. Apply factorisation to a realistic situation. What would you do or calculate? Justify your response.
40. Apply equations and real-life modelling. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Algebraic expressions. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of identities. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of factorisation. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of equations and real-life modelling.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Algebraic expressions. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of identities. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of factorisation. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of equations and real-life modelling.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Algebraic expressions. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of identities. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show factorisation. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show equations and real-life modelling.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Algebraic expressions. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show identities. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show factorisation. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show equations and real-life modelling.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Algebraic expressions. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show identities. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show factorisation. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show equations and real-life modelling.. Include the relevant rule or principle.

MATHEMATICS — 14. Area

Coverage: ■ Area of triangles/parallelograms/trapeziums/composite figures; perimeter; measurement reasoning.

Understanding

1. What is ■ Area of triangles/parallelograms/trapeziums/composite figures, and what are its main features? Use a simple example.
2. What is perimeter, and what are its main features? Use a school-life example.
3. What is measurement reasoning., and what are its main features? Use a real-world example.
4. What is ■ Area of triangles/parallelograms/trapeziums/composite figures, and what are its main features? Show the idea with a diagram where appropriate.
5. What is perimeter, and what are its main features? Explain in your own words.
6. What is measurement reasoning., and what are its main features? Give a step-by-step response.
7. What is ■ Area of triangles/parallelograms/trapeziums/composite figures, and what are its main features? Mention one common misconception.
8. What is perimeter, and what are its main features? State one practical application.
9. What is measurement reasoning., and what are its main features? Justify your response.
10. What is ■ Area of triangles/parallelograms/trapeziums/composite figures, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to perimeter and explain each. Use a simple example.
12. Give two examples related to measurement reasoning. and explain each. Use a school-life example.
13. Give two examples related to ■ Area of triangles/parallelograms/trapeziums/composite figures and explain each. Use a real-world example.
14. Give two examples related to perimeter and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to measurement reasoning. and explain each. Explain in your own words.
16. Give two examples related to ■ Area of triangles/parallelograms/trapeziums/composite figures and explain each. Give a step-by-step response.
17. Give two examples related to perimeter and explain each. Mention one common misconception.
18. Give two examples related to measurement reasoning. and explain each. State one practical application.
19. Give two examples related to ■ Area of triangles/parallelograms/trapeziums/composite figures and explain each. Justify your response.
20. Give two examples related to perimeter and explain each. Include the relevant rule or principle.

Comparison

21. How is measurement reasoning. related to ■ Area of triangles/parallelograms/trapeziums/composite figures? Compare the two ideas. Use a simple example.
22. How is ■ Area of triangles/parallelograms/trapeziums/composite figures related to perimeter? Compare the two ideas. Use a school-life example.
23. How is perimeter related to measurement reasoning.? Compare the two ideas. Use a real-world example.
24. How is measurement reasoning. related to ■ Area of triangles/parallelograms/trapeziums/composite figures? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Area of triangles/parallelograms/trapeziums/composite figures related to perimeter? Compare the two ideas. Explain in your own words.
26. How is perimeter related to measurement reasoning.? Compare the two ideas. Give a step-by-step response.
27. How is measurement reasoning. related to ■ Area of triangles/parallelograms/trapeziums/composite figures? Compare the two ideas. Mention one common misconception.
28. How is ■ Area of triangles/parallelograms/trapeziums/composite figures related to perimeter? Compare the two ideas. State one practical application.
29. How is perimeter related to measurement reasoning.? Compare the two ideas. Justify your response.
30. How is measurement reasoning. related to ■ Area of triangles/parallelograms/trapeziums/composite figures? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Area of triangles/parallelograms/trapeziums/composite figures to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply perimeter to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply measurement reasoning. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Area of triangles/parallelograms/trapeziums/composite figures to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply perimeter to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply measurement reasoning. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Area of triangles/parallelograms/trapeziums/composite figures to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply perimeter to a realistic situation. What would you do or calculate? State one practical application.
39. Apply measurement reasoning. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Area of triangles/parallelograms/trapeziums/composite figures to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of perimeter. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of measurement reasoning.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Area of triangles/parallelograms/trapeziums/composite figures. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of perimeter. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of measurement reasoning.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Area of triangles/parallelograms/trapeziums/composite figures. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of perimeter. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of measurement reasoning.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Area of triangles/parallelograms/trapeziums/composite figures. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of perimeter. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show measurement reasoning.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area of triangles/parallelograms/trapeziums/composite figures. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show perimeter. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show measurement reasoning.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area of triangles/parallelograms/trapeziums/composite figures. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show perimeter. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show measurement reasoning.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Area of triangles/parallelograms/trapeziums/composite figures. State one practical application.

59. Design a short activity, investigation, problem, model, or demonstration to test or show perimeter. Justify your response.

60. Design a short activity, investigation, problem, model, or demonstration to test or show measurement reasoning.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 1. Exploring the Investigative World of Science

Coverage: ■ Observation; questioning; hypothesis; experimental design; evidence and conclusions.

Understanding

1. What is ■ Observation, and what are its main features? Use a simple example.
2. What is questioning, and what are its main features? Use a school-life example.
3. What is hypothesis, and what are its main features? Use a real-world example.
4. What is experimental design, and what are its main features? Show the idea with a diagram where appropriate.
5. What is evidence and conclusions., and what are its main features? Explain in your own words.
6. What is ■ Observation, and what are its main features? Give a step-by-step response.
7. What is questioning, and what are its main features? Mention one common misconception.
8. What is hypothesis, and what are its main features? State one practical application.
9. What is experimental design, and what are its main features? Justify your response.
10. What is evidence and conclusions., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Observation and explain each. Use a simple example.
12. Give two examples related to questioning and explain each. Use a school-life example.
13. Give two examples related to hypothesis and explain each. Use a real-world example.
14. Give two examples related to experimental design and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to evidence and conclusions. and explain each. Explain in your own words.
16. Give two examples related to ■ Observation and explain each. Give a step-by-step response.
17. Give two examples related to questioning and explain each. Mention one common misconception.
18. Give two examples related to hypothesis and explain each. State one practical application.
19. Give two examples related to experimental design and explain each. Justify your response.
20. Give two examples related to evidence and conclusions. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Observation related to questioning? Compare the two ideas. Use a simple example.
22. How is questioning related to hypothesis? Compare the two ideas. Use a school-life example.
23. How is hypothesis related to experimental design? Compare the two ideas. Use a real-world example.
24. How is experimental design related to evidence and conclusions.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is evidence and conclusions. related to ■ Observation? Compare the two ideas. Explain in your own words.
26. How is ■ Observation related to questioning? Compare the two ideas. Give a step-by-step response.
27. How is questioning related to hypothesis? Compare the two ideas. Mention one common misconception.
28. How is hypothesis related to experimental design? Compare the two ideas. State one practical application.
29. How is experimental design related to evidence and conclusions.? Compare the two ideas. Justify your response.
30. How is evidence and conclusions. related to ■ Observation? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Observation to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply questioning to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply hypothesis to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply experimental design to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply evidence and conclusions. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Observation to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply questioning to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply hypothesis to a realistic situation. What would you do or calculate? State one practical application.
39. Apply experimental design to a realistic situation. What would you do or calculate? Justify your response.
40. Apply evidence and conclusions. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Observation. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of questioning. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of hypothesis. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of experimental design. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of evidence and conclusions.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Observation. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of questioning. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of hypothesis. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of experimental design. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of evidence and conclusions.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Observation. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show questioning. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show hypothesis. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show experimental design. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show evidence and conclusions.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Observation. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show questioning. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show hypothesis. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show experimental design. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show evidence and conclusions.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 2. The Invisible Living World: Beyond Our Naked Eye

Coverage: ■ Microscopy; microorganisms; cells; useful/harmful microbes.

Understanding

1. What is ■ Microscopy, and what are its main features? Use a simple example.
2. What is microorganisms, and what are its main features? Use a school-life example.
3. What is cells, and what are its main features? Use a real-world example.
4. What is useful/harmful microbes., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Microscopy, and what are its main features? Explain in your own words.
6. What is microorganisms, and what are its main features? Give a step-by-step response.
7. What is cells, and what are its main features? Mention one common misconception.
8. What is useful/harmful microbes., and what are its main features? State one practical application.
9. What is ■ Microscopy, and what are its main features? Justify your response.
10. What is microorganisms, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to cells and explain each. Use a simple example.
12. Give two examples related to useful/harmful microbes. and explain each. Use a school-life example.
13. Give two examples related to ■ Microscopy and explain each. Use a real-world example.
14. Give two examples related to microorganisms and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to cells and explain each. Explain in your own words.
16. Give two examples related to useful/harmful microbes. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Microscopy and explain each. Mention one common misconception.
18. Give two examples related to microorganisms and explain each. State one practical application.
19. Give two examples related to cells and explain each. Justify your response.
20. Give two examples related to useful/harmful microbes. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Microscopy related to microorganisms? Compare the two ideas. Use a simple example.
22. How is microorganisms related to cells? Compare the two ideas. Use a school-life example.
23. How is cells related to useful/harmful microbes.? Compare the two ideas. Use a real-world example.
24. How is useful/harmful microbes. related to ■ Microscopy? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Microscopy related to microorganisms? Compare the two ideas. Explain in your own words.
26. How is microorganisms related to cells? Compare the two ideas. Give a step-by-step response.
27. How is cells related to useful/harmful microbes.? Compare the two ideas. Mention one common misconception.
28. How is useful/harmful microbes. related to ■ Microscopy? Compare the two ideas. State one practical application.
29. How is ■ Microscopy related to microorganisms? Compare the two ideas. Justify your response.
30. How is microorganisms related to cells? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply cells to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply useful/harmful microbes. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Microscopy to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply microorganisms to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply cells to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply useful/harmful microbes. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Microscopy to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply microorganisms to a realistic situation. What would you do or calculate? State one practical application.
39. Apply cells to a realistic situation. What would you do or calculate? Justify your response.
40. Apply useful/harmful microbes. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Microscopy. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of microorganisms. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of cells. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of useful/harmful microbes.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Microscopy. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of microorganisms. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of cells. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of useful/harmful microbes.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Microscopy. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of microorganisms. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show cells. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show useful/harmful microbes.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Microscopy. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show microorganisms. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show cells. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show useful/harmful microbes.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Microscopy. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show microorganisms. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show cells. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show useful/harmful microbes.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 3. Health: The Ultimate Treasure

Coverage: ■ Health; nutrition; disease; immunity; hygiene; prevention.

Understanding

1. What is ■ Health, and what are its main features? Use a simple example.
2. What is nutrition, and what are its main features? Use a school-life example.
3. What is disease, and what are its main features? Use a real-world example.
4. What is immunity, and what are its main features? Show the idea with a diagram where appropriate.
5. What is hygiene, and what are its main features? Explain in your own words.
6. What is prevention., and what are its main features? Give a step-by-step response.
7. What is ■ Health, and what are its main features? Mention one common misconception.
8. What is nutrition, and what are its main features? State one practical application.
9. What is disease, and what are its main features? Justify your response.
10. What is immunity, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to hygiene and explain each. Use a simple example.
12. Give two examples related to prevention. and explain each. Use a school-life example.
13. Give two examples related to ■ Health and explain each. Use a real-world example.
14. Give two examples related to nutrition and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to disease and explain each. Explain in your own words.
16. Give two examples related to immunity and explain each. Give a step-by-step response.
17. Give two examples related to hygiene and explain each. Mention one common misconception.
18. Give two examples related to prevention. and explain each. State one practical application.
19. Give two examples related to ■ Health and explain each. Justify your response.
20. Give two examples related to nutrition and explain each. Include the relevant rule or principle.

Comparison

21. How is disease related to immunity? Compare the two ideas. Use a simple example.
22. How is immunity related to hygiene? Compare the two ideas. Use a school-life example.
23. How is hygiene related to prevention.? Compare the two ideas. Use a real-world example.
24. How is prevention. related to ■ Health? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Health related to nutrition? Compare the two ideas. Explain in your own words.
26. How is nutrition related to disease? Compare the two ideas. Give a step-by-step response.
27. How is disease related to immunity? Compare the two ideas. Mention one common misconception.
28. How is immunity related to hygiene? Compare the two ideas. State one practical application.
29. How is hygiene related to prevention.? Compare the two ideas. Justify your response.
30. How is prevention. related to ■ Health? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Health to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply nutrition to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply disease to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply immunity to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply hygiene to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply prevention. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Health to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply nutrition to a realistic situation. What would you do or calculate? State one practical application.
39. Apply disease to a realistic situation. What would you do or calculate? Justify your response.
40. Apply immunity to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of hygiene. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of prevention.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Health. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of nutrition. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of disease. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of immunity. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of hygiene. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of prevention.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Health. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of nutrition. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show disease. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show immunity. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show hygiene. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show prevention.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Health. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show nutrition. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show disease. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show immunity. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show hygiene. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show prevention.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 4. Electricity: Magnetic and Heating Effects

Coverage: NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus

Understanding

1. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a simple example.
2. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a school-life example.
3. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Use a real-world example.
4. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Show the idea with a diagram where appropriate.
5. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Explain in your own words.
6. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Give a step-by-step response.
7. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Mention one common misconception.
8. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? State one practical application.
9. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Justify your response.
10. What is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a simple example.
12. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a school-life example.
13. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Use a real-world example.
14. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Explain in your own words.
16. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Give a step-by-step response.
17. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Mention one common misconception.
18. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. State one practical application.
19. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Justify your response.
20. Give two examples related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus and explain each. Include the relevant rule or principle.

Comparison

21. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a simple example.
22. How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a school-life example.

- 23.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Use a real-world example.
- 24.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Show the idea with a diagram where appropriate.
- 25.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Explain in your own words.
- 26.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Give a step-by-step response.
- 27.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Mention one common misconception.
- 28.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. State one practical application.
- 29.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Justify your response.
- 30.** How is NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus related to NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus? Compare the two ideas. Include the relevant rule or principle.

Application

- 31.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a simple example.
- 32.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a school-life example.
- 33.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Use a real-world example.
- 34.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
- 35.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Explain in your own words.
- 36.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Give a step-by-step response.
- 37.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Mention one common misconception.
- 38.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? State one practical application.
- 39.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Justify your response.
- 40.** Apply NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

- 41.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a simple example.
- 42.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a school-life example.
- 43.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Use a real-world example.
- 44.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
- 45.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Explain in your own words.
- 46.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Give a step-by-step response.
- 47.** A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Mention one common misconception.

48. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show NCERT Curriculum Map ■ 2026–27 ■ Supplementary skill-gap items are not official syllabus. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 5. Exploring Forces

Coverage: ■ Contact/non-contact forces; friction; effects of force; motion.

Understanding

1. What is ■ Contact/non-contact forces, and what are its main features? Use a simple example.
2. What is friction, and what are its main features? Use a school-life example.
3. What is effects of force, and what are its main features? Use a real-world example.
4. What is motion., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Contact/non-contact forces, and what are its main features? Explain in your own words.
6. What is friction, and what are its main features? Give a step-by-step response.
7. What is effects of force, and what are its main features? Mention one common misconception.
8. What is motion., and what are its main features? State one practical application.
9. What is ■ Contact/non-contact forces, and what are its main features? Justify your response.
10. What is friction, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to effects of force and explain each. Use a simple example.
12. Give two examples related to motion. and explain each. Use a school-life example.
13. Give two examples related to ■ Contact/non-contact forces and explain each. Use a real-world example.
14. Give two examples related to friction and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to effects of force and explain each. Explain in your own words.
16. Give two examples related to motion. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Contact/non-contact forces and explain each. Mention one common misconception.
18. Give two examples related to friction and explain each. State one practical application.
19. Give two examples related to effects of force and explain each. Justify your response.
20. Give two examples related to motion. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Contact/non-contact forces related to friction? Compare the two ideas. Use a simple example.
22. How is friction related to effects of force? Compare the two ideas. Use a school-life example.
23. How is effects of force related to motion.? Compare the two ideas. Use a real-world example.
24. How is motion. related to ■ Contact/non-contact forces? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Contact/non-contact forces related to friction? Compare the two ideas. Explain in your own words.
26. How is friction related to effects of force? Compare the two ideas. Give a step-by-step response.
27. How is effects of force related to motion.? Compare the two ideas. Mention one common misconception.
28. How is motion. related to ■ Contact/non-contact forces? Compare the two ideas. State one practical application.
29. How is ■ Contact/non-contact forces related to friction? Compare the two ideas. Justify your response.
30. How is friction related to effects of force? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply effects of force to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply motion. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Contact/non-contact forces to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply friction to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply effects of force to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply motion. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Contact/non-contact forces to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply friction to a realistic situation. What would you do or calculate? State one practical application.
39. Apply effects of force to a realistic situation. What would you do or calculate? Justify your response.
40. Apply motion. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Contact/non-contact forces. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of friction. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of effects of force. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of motion.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Contact/non-contact forces. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of friction. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of effects of force. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of motion.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Contact/non-contact forces. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of friction. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show effects of force. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show motion.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Contact/non-contact forces. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show friction. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show effects of force. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show motion.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Contact/non-contact forces. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show friction. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show effects of force. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show motion.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 6. Pressure, Winds, Storms, and Cyclones

Coverage: ■ Pressure; air pressure; wind formation; storms; thunderstorms; lightning; cyclones; safety.

Understanding

1. What is ■ Pressure, and what are its main features? Use a simple example.
2. What is air pressure, and what are its main features? Use a school-life example.
3. What is wind formation, and what are its main features? Use a real-world example.
4. What is storms, and what are its main features? Show the idea with a diagram where appropriate.
5. What is thunderstorms, and what are its main features? Explain in your own words.
6. What is lightning, and what are its main features? Give a step-by-step response.
7. What is cyclones, and what are its main features? Mention one common misconception.
8. What is safety., and what are its main features? State one practical application.
9. What is ■ Pressure, and what are its main features? Justify your response.
10. What is air pressure, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to wind formation and explain each. Use a simple example.
12. Give two examples related to storms and explain each. Use a school-life example.
13. Give two examples related to thunderstorms and explain each. Use a real-world example.
14. Give two examples related to lightning and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to cyclones and explain each. Explain in your own words.
16. Give two examples related to safety. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Pressure and explain each. Mention one common misconception.
18. Give two examples related to air pressure and explain each. State one practical application.
19. Give two examples related to wind formation and explain each. Justify your response.
20. Give two examples related to storms and explain each. Include the relevant rule or principle.

Comparison

21. How is thunderstorms related to lightning? Compare the two ideas. Use a simple example.
22. How is lightning related to cyclones? Compare the two ideas. Use a school-life example.
23. How is cyclones related to safety.? Compare the two ideas. Use a real-world example.
24. How is safety. related to ■ Pressure? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Pressure related to air pressure? Compare the two ideas. Explain in your own words.
26. How is air pressure related to wind formation? Compare the two ideas. Give a step-by-step response.
27. How is wind formation related to storms? Compare the two ideas. Mention one common misconception.
28. How is storms related to thunderstorms? Compare the two ideas. State one practical application.
29. How is thunderstorms related to lightning? Compare the two ideas. Justify your response.
30. How is lightning related to cyclones? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply cyclones to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply safety. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Pressure to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply air pressure to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply wind formation to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply storms to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply thunderstorms to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply lightning to a realistic situation. What would you do or calculate? State one practical application.
39. Apply cyclones to a realistic situation. What would you do or calculate? Justify your response.
40. Apply safety. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Pressure. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of air pressure. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of wind formation. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of storms. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of thunderstorms. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of lightning. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of cyclones. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of safety.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Pressure. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of air pressure. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show wind formation. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show storms. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show thunderstorms. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show lightning. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show cyclones. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show safety.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Pressure. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show air pressure. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show wind formation. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show storms. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 7. Particulate Nature of Matter

Coverage: ■ Particle model; arrangement/motion of particles; states and observable properties.

Understanding

1. What is ■ Particle model, and what are its main features? Use a simple example.
2. What is arrangement/motion of particles, and what are its main features? Use a school-life example.
3. What is states and observable properties., and what are its main features? Use a real-world example.
4. What is ■ Particle model, and what are its main features? Show the idea with a diagram where appropriate.
5. What is arrangement/motion of particles, and what are its main features? Explain in your own words.
6. What is states and observable properties., and what are its main features? Give a step-by-step response.
7. What is ■ Particle model, and what are its main features? Mention one common misconception.
8. What is arrangement/motion of particles, and what are its main features? State one practical application.
9. What is states and observable properties., and what are its main features? Justify your response.
10. What is ■ Particle model, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to arrangement/motion of particles and explain each. Use a simple example.
12. Give two examples related to states and observable properties. and explain each. Use a school-life example.
13. Give two examples related to ■ Particle model and explain each. Use a real-world example.
14. Give two examples related to arrangement/motion of particles and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to states and observable properties. and explain each. Explain in your own words.
16. Give two examples related to ■ Particle model and explain each. Give a step-by-step response.
17. Give two examples related to arrangement/motion of particles and explain each. Mention one common misconception.
18. Give two examples related to states and observable properties. and explain each. State one practical application.
19. Give two examples related to ■ Particle model and explain each. Justify your response.
20. Give two examples related to arrangement/motion of particles and explain each. Include the relevant rule or principle.

Comparison

21. How is states and observable properties. related to ■ Particle model? Compare the two ideas. Use a simple example.
22. How is ■ Particle model related to arrangement/motion of particles? Compare the two ideas. Use a school-life example.
23. How is arrangement/motion of particles related to states and observable properties.? Compare the two ideas. Use a real-world example.
24. How is states and observable properties. related to ■ Particle model? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Particle model related to arrangement/motion of particles? Compare the two ideas. Explain in your own words.
26. How is arrangement/motion of particles related to states and observable properties.? Compare the two ideas. Give a step-by-step response.
27. How is states and observable properties. related to ■ Particle model? Compare the two ideas. Mention one common misconception.
28. How is ■ Particle model related to arrangement/motion of particles? Compare the two ideas. State one practical application.
29. How is arrangement/motion of particles related to states and observable properties.? Compare the two ideas. Justify your response.
30. How is states and observable properties. related to ■ Particle model? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Particle model to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply arrangement/motion of particles to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply states and observable properties. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Particle model to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply arrangement/motion of particles to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply states and observable properties. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Particle model to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply arrangement/motion of particles to a realistic situation. What would you do or calculate? State one practical application.
39. Apply states and observable properties. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Particle model to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of arrangement/motion of particles. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of states and observable properties.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Particle model. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of arrangement/motion of particles. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of states and observable properties.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Particle model. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of arrangement/motion of particles. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of states and observable properties.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Particle model. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of arrangement/motion of particles. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show states and observable properties.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Particle model. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show arrangement/motion of particles. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show states and observable properties.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Particle model. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show arrangement/motion of particles. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show states and observable properties.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Particle model. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show arrangement/motion of particles. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show states and observable properties.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 8. Nature of Matter: Elements, Compounds, and Mixtures

Coverage: ■ Elements; compounds; mixtures; physical/chemical distinctions; separation.

Understanding

1. What is ■ Elements, and what are its main features? Use a simple example.
2. What is compounds, and what are its main features? Use a school-life example.
3. What is mixtures, and what are its main features? Use a real-world example.
4. What is physical/chemical distinctions, and what are its main features? Show the idea with a diagram where appropriate.
5. What is separation., and what are its main features? Explain in your own words.
6. What is ■ Elements, and what are its main features? Give a step-by-step response.
7. What is compounds, and what are its main features? Mention one common misconception.
8. What is mixtures, and what are its main features? State one practical application.
9. What is physical/chemical distinctions, and what are its main features? Justify your response.
10. What is separation., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Elements and explain each. Use a simple example.
12. Give two examples related to compounds and explain each. Use a school-life example.
13. Give two examples related to mixtures and explain each. Use a real-world example.
14. Give two examples related to physical/chemical distinctions and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to separation. and explain each. Explain in your own words.
16. Give two examples related to ■ Elements and explain each. Give a step-by-step response.
17. Give two examples related to compounds and explain each. Mention one common misconception.
18. Give two examples related to mixtures and explain each. State one practical application.
19. Give two examples related to physical/chemical distinctions and explain each. Justify your response.
20. Give two examples related to separation. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Elements related to compounds? Compare the two ideas. Use a simple example.
22. How is compounds related to mixtures? Compare the two ideas. Use a school-life example.
23. How is mixtures related to physical/chemical distinctions? Compare the two ideas. Use a real-world example.
24. How is physical/chemical distinctions related to separation.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is separation. related to ■ Elements? Compare the two ideas. Explain in your own words.
26. How is ■ Elements related to compounds? Compare the two ideas. Give a step-by-step response.
27. How is compounds related to mixtures? Compare the two ideas. Mention one common misconception.
28. How is mixtures related to physical/chemical distinctions? Compare the two ideas. State one practical application.
29. How is physical/chemical distinctions related to separation.? Compare the two ideas. Justify your response.
30. How is separation. related to ■ Elements? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Elements to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply compounds to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply mixtures to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply physical/chemical distinctions to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply separation. to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply ■ Elements to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply compounds to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply mixtures to a realistic situation. What would you do or calculate? State one practical application.
39. Apply physical/chemical distinctions to a realistic situation. What would you do or calculate? Justify your response.
40. Apply separation. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Elements. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of compounds. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of mixtures. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of physical/chemical distinctions. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of separation.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Elements. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of compounds. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of mixtures. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of physical/chemical distinctions. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of separation.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Elements. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show compounds. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show mixtures. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show physical/chemical distinctions. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show separation.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Elements. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show compounds. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show mixtures. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show physical/chemical distinctions. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show separation.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 9. The Amazing World of Solutes, Solvents, and Solutions

Coverage: ■ Solute/solvent/solution; concentration; solubility; everyday solutions.

Understanding

1. What is ■ Solute/solvent/solution, and what are its main features? Use a simple example.
2. What is concentration, and what are its main features? Use a school-life example.
3. What is solubility, and what are its main features? Use a real-world example.
4. What is everyday solutions., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Solute/solvent/solution, and what are its main features? Explain in your own words.
6. What is concentration, and what are its main features? Give a step-by-step response.
7. What is solubility, and what are its main features? Mention one common misconception.
8. What is everyday solutions., and what are its main features? State one practical application.
9. What is ■ Solute/solvent/solution, and what are its main features? Justify your response.
10. What is concentration, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to solubility and explain each. Use a simple example.
12. Give two examples related to everyday solutions. and explain each. Use a school-life example.
13. Give two examples related to ■ Solute/solvent/solution and explain each. Use a real-world example.
14. Give two examples related to concentration and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to solubility and explain each. Explain in your own words.
16. Give two examples related to everyday solutions. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Solute/solvent/solution and explain each. Mention one common misconception.
18. Give two examples related to concentration and explain each. State one practical application.
19. Give two examples related to solubility and explain each. Justify your response.
20. Give two examples related to everyday solutions. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Solute/solvent/solution related to concentration? Compare the two ideas. Use a simple example.
22. How is concentration related to solubility? Compare the two ideas. Use a school-life example.
23. How is solubility related to everyday solutions.? Compare the two ideas. Use a real-world example.
24. How is everyday solutions. related to ■ Solute/solvent/solution? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Solute/solvent/solution related to concentration? Compare the two ideas. Explain in your own words.
26. How is concentration related to solubility? Compare the two ideas. Give a step-by-step response.
27. How is solubility related to everyday solutions.? Compare the two ideas. Mention one common misconception.
28. How is everyday solutions. related to ■ Solute/solvent/solution? Compare the two ideas. State one practical application.
29. How is ■ Solute/solvent/solution related to concentration? Compare the two ideas. Justify your response.
30. How is concentration related to solubility? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply solubility to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply everyday solutions. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Solute/solvent/solution to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply concentration to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply solubility to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply everyday solutions. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Solute/solvent/solution to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply concentration to a realistic situation. What would you do or calculate? State one practical application.
39. Apply solubility to a realistic situation. What would you do or calculate? Justify your response.
40. Apply everyday solutions. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Solute/solvent/solution. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of concentration. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of solubility. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of everyday solutions.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Solute/solvent/solution. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of concentration. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of solubility. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of everyday solutions.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Solute/solvent/solution. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of concentration. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show solubility. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show everyday solutions.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Solute/solvent/solution. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show concentration. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show solubility. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show everyday solutions.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Solute/solvent/solution. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show concentration. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show solubility. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show everyday solutions.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 10. Light: Mirrors and Lenses

Coverage: ■ Reflection; mirrors; refraction; lenses; image formation and applications.

Understanding

1. What is ■ Reflection, and what are its main features? Use a simple example.
2. What is mirrors, and what are its main features? Use a school-life example.
3. What is refraction, and what are its main features? Use a real-world example.
4. What is lenses, and what are its main features? Show the idea with a diagram where appropriate.
5. What is image formation and applications., and what are its main features? Explain in your own words.
6. What is ■ Reflection, and what are its main features? Give a step-by-step response.
7. What is mirrors, and what are its main features? Mention one common misconception.
8. What is refraction, and what are its main features? State one practical application.
9. What is lenses, and what are its main features? Justify your response.
10. What is image formation and applications., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Reflection and explain each. Use a simple example.
12. Give two examples related to mirrors and explain each. Use a school-life example.
13. Give two examples related to refraction and explain each. Use a real-world example.
14. Give two examples related to lenses and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to image formation and applications. and explain each. Explain in your own words.
16. Give two examples related to ■ Reflection and explain each. Give a step-by-step response.
17. Give two examples related to mirrors and explain each. Mention one common misconception.
18. Give two examples related to refraction and explain each. State one practical application.
19. Give two examples related to lenses and explain each. Justify your response.
20. Give two examples related to image formation and applications. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Reflection related to mirrors? Compare the two ideas. Use a simple example.
22. How is mirrors related to refraction? Compare the two ideas. Use a school-life example.
23. How is refraction related to lenses? Compare the two ideas. Use a real-world example.
24. How is lenses related to image formation and applications.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is image formation and applications. related to ■ Reflection? Compare the two ideas. Explain in your own words.
26. How is ■ Reflection related to mirrors? Compare the two ideas. Give a step-by-step response.
27. How is mirrors related to refraction? Compare the two ideas. Mention one common misconception.
28. How is refraction related to lenses? Compare the two ideas. State one practical application.
29. How is lenses related to image formation and applications.? Compare the two ideas. Justify your response.
30. How is image formation and applications. related to ■ Reflection? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Reflection to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply mirrors to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply refraction to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply lenses to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply image formation and applications. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Reflection to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply mirrors to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply refraction to a realistic situation. What would you do or calculate? State one practical application.
39. Apply lenses to a realistic situation. What would you do or calculate? Justify your response.
40. Apply image formation and applications. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Reflection. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of mirrors. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of refraction. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of lenses. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of image formation and applications.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Reflection. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of mirrors. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of refraction. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of lenses. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of image formation and applications.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Reflection. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show mirrors. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show refraction. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show lenses. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show image formation and applications.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Reflection. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show mirrors. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show refraction. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show lenses. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show image formation and applications.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 11. Keeping Time with the Skies

Coverage: ■ Sky observations; apparent motion; timekeeping; celestial patterns.

Understanding

1. What is ■ Sky observations, and what are its main features? Use a simple example.
2. What is apparent motion, and what are its main features? Use a school-life example.
3. What is timekeeping, and what are its main features? Use a real-world example.
4. What is celestial patterns., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Sky observations, and what are its main features? Explain in your own words.
6. What is apparent motion, and what are its main features? Give a step-by-step response.
7. What is timekeeping, and what are its main features? Mention one common misconception.
8. What is celestial patterns., and what are its main features? State one practical application.
9. What is ■ Sky observations, and what are its main features? Justify your response.
10. What is apparent motion, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to timekeeping and explain each. Use a simple example.
12. Give two examples related to celestial patterns. and explain each. Use a school-life example.
13. Give two examples related to ■ Sky observations and explain each. Use a real-world example.
14. Give two examples related to apparent motion and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to timekeeping and explain each. Explain in your own words.
16. Give two examples related to celestial patterns. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Sky observations and explain each. Mention one common misconception.
18. Give two examples related to apparent motion and explain each. State one practical application.
19. Give two examples related to timekeeping and explain each. Justify your response.
20. Give two examples related to celestial patterns. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Sky observations related to apparent motion? Compare the two ideas. Use a simple example.
22. How is apparent motion related to timekeeping? Compare the two ideas. Use a school-life example.
23. How is timekeeping related to celestial patterns.? Compare the two ideas. Use a real-world example.
24. How is celestial patterns. related to ■ Sky observations? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Sky observations related to apparent motion? Compare the two ideas. Explain in your own words.
26. How is apparent motion related to timekeeping? Compare the two ideas. Give a step-by-step response.
27. How is timekeeping related to celestial patterns.? Compare the two ideas. Mention one common misconception.
28. How is celestial patterns. related to ■ Sky observations? Compare the two ideas. State one practical application.
29. How is ■ Sky observations related to apparent motion? Compare the two ideas. Justify your response.
30. How is apparent motion related to timekeeping? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply timekeeping to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply celestial patterns. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Sky observations to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply apparent motion to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply timekeeping to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply celestial patterns. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Sky observations to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply apparent motion to a realistic situation. What would you do or calculate? State one practical application.
39. Apply timekeeping to a realistic situation. What would you do or calculate? Justify your response.
40. Apply celestial patterns. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Sky observations. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of apparent motion. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of timekeeping. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of celestial patterns.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Sky observations. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of apparent motion. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of timekeeping. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of celestial patterns.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Sky observations. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of apparent motion. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show timekeeping. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show celestial patterns.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Sky observations. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show apparent motion. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show timekeeping. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show celestial patterns.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Sky observations. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show apparent motion. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show timekeeping. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show celestial patterns.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 12. How Nature Works in Harmony

Coverage: ■ Interdependence; ecosystems; cycles; balance and human impact.

Understanding

1. What is ■ Interdependence, and what are its main features? Use a simple example.
2. What is ecosystems, and what are its main features? Use a school-life example.
3. What is cycles, and what are its main features? Use a real-world example.
4. What is balance and human impact., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Interdependence, and what are its main features? Explain in your own words.
6. What is ecosystems, and what are its main features? Give a step-by-step response.
7. What is cycles, and what are its main features? Mention one common misconception.
8. What is balance and human impact., and what are its main features? State one practical application.
9. What is ■ Interdependence, and what are its main features? Justify your response.
10. What is ecosystems, and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to cycles and explain each. Use a simple example.
12. Give two examples related to balance and human impact. and explain each. Use a school-life example.
13. Give two examples related to ■ Interdependence and explain each. Use a real-world example.
14. Give two examples related to ecosystems and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to cycles and explain each. Explain in your own words.
16. Give two examples related to balance and human impact. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Interdependence and explain each. Mention one common misconception.
18. Give two examples related to ecosystems and explain each. State one practical application.
19. Give two examples related to cycles and explain each. Justify your response.
20. Give two examples related to balance and human impact. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Interdependence related to ecosystems? Compare the two ideas. Use a simple example.
22. How is ecosystems related to cycles? Compare the two ideas. Use a school-life example.
23. How is cycles related to balance and human impact.? Compare the two ideas. Use a real-world example.
24. How is balance and human impact. related to ■ Interdependence? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Interdependence related to ecosystems? Compare the two ideas. Explain in your own words.
26. How is ecosystems related to cycles? Compare the two ideas. Give a step-by-step response.
27. How is cycles related to balance and human impact.? Compare the two ideas. Mention one common misconception.
28. How is balance and human impact. related to ■ Interdependence? Compare the two ideas. State one practical application.
29. How is ■ Interdependence related to ecosystems? Compare the two ideas. Justify your response.
30. How is ecosystems related to cycles? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply cycles to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply balance and human impact. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Interdependence to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ecosystems to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply cycles to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply balance and human impact. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Interdependence to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply ecosystems to a realistic situation. What would you do or calculate? State one practical application.
39. Apply cycles to a realistic situation. What would you do or calculate? Justify your response.
40. Apply balance and human impact. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Interdependence. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of ecosystems. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of cycles. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of balance and human impact.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Interdependence. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ecosystems. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of cycles. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of balance and human impact.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Interdependence. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of ecosystems. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show cycles. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show balance and human impact.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Interdependence. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show ecosystems. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show cycles. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show balance and human impact.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Interdependence. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ecosystems. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show cycles. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show balance and human impact.. Include the relevant rule or principle.

SCIENCE / INTEGRATED SCIENCE — 13. Our Home: Earth, a Unique Life Sustaining Planet

Coverage: ■ Earth systems; resources; atmosphere; life-supporting conditions; sustainability.

Understanding

1. What is ■ Earth systems, and what are its main features? Use a simple example.
2. What is resources, and what are its main features? Use a school-life example.
3. What is atmosphere, and what are its main features? Use a real-world example.
4. What is life-supporting conditions, and what are its main features? Show the idea with a diagram where appropriate.
5. What is sustainability., and what are its main features? Explain in your own words.
6. What is ■ Earth systems, and what are its main features? Give a step-by-step response.
7. What is resources, and what are its main features? Mention one common misconception.
8. What is atmosphere, and what are its main features? State one practical application.
9. What is life-supporting conditions, and what are its main features? Justify your response.
10. What is sustainability., and what are its main features? Include the relevant rule or principle.

Examples & Classification

11. Give two examples related to ■ Earth systems and explain each. Use a simple example.
12. Give two examples related to resources and explain each. Use a school-life example.
13. Give two examples related to atmosphere and explain each. Use a real-world example.
14. Give two examples related to life-supporting conditions and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to sustainability. and explain each. Explain in your own words.
16. Give two examples related to ■ Earth systems and explain each. Give a step-by-step response.
17. Give two examples related to resources and explain each. Mention one common misconception.
18. Give two examples related to atmosphere and explain each. State one practical application.
19. Give two examples related to life-supporting conditions and explain each. Justify your response.
20. Give two examples related to sustainability. and explain each. Include the relevant rule or principle.

Comparison

21. How is ■ Earth systems related to resources? Compare the two ideas. Use a simple example.
22. How is resources related to atmosphere? Compare the two ideas. Use a school-life example.
23. How is atmosphere related to life-supporting conditions? Compare the two ideas. Use a real-world example.
24. How is life-supporting conditions related to sustainability.? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is sustainability. related to ■ Earth systems? Compare the two ideas. Explain in your own words.
26. How is ■ Earth systems related to resources? Compare the two ideas. Give a step-by-step response.
27. How is resources related to atmosphere? Compare the two ideas. Mention one common misconception.
28. How is atmosphere related to life-supporting conditions? Compare the two ideas. State one practical application.
29. How is life-supporting conditions related to sustainability.? Compare the two ideas. Justify your response.
30. How is sustainability. related to ■ Earth systems? Compare the two ideas. Include the relevant rule or principle.

Application

31. Apply ■ Earth systems to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply resources to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply atmosphere to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply life-supporting conditions to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply sustainability. to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply ■ Earth systems to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply resources to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply atmosphere to a realistic situation. What would you do or calculate? State one practical application.
39. Apply life-supporting conditions to a realistic situation. What would you do or calculate? Justify your response.
40. Apply sustainability. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Earth systems. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of resources. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of atmosphere. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of life-supporting conditions. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of sustainability.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Earth systems. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of resources. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of atmosphere. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of life-supporting conditions. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of sustainability.. Identify a likely error and explain the correction. Include the relevant rule or principle.

Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Earth systems. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show resources. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show atmosphere. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show life-supporting conditions. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show sustainability.. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Earth systems. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show resources. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show atmosphere. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show life-supporting conditions. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show sustainability.. Include the relevant rule or principle.